OBJECT ORIENTED SOFTWARE ENGINEERING

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OBJECT ORIENTED SOFTWARE ENGINEERING

3. Behavioral Models
(a) State Models
(b) Sequence Models

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OBJECT ORIENTED SOFTWARE ENGINEERING

(a) State Model

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- State model describes the **sequence of operations** **that occur in response to** **external stimuli.**

- The state model consists of **multiple state diagrams**, **one for each class with temporal behaviour.**

- It express the state diagram for graphical representation for state machine that relates **events** and **states**.
    - Events represents **external Stimuli**
    - State represents **values of objects**.

Events

- An Event is -- an **occurrence at a point in time.**

- By definition, an event happens instantaneously with regard to the time scale of an application.

  - **One event may logically precede or follow another(related)** or

  - **Two events may be unrelated(concurrent).**

Events…

Events for example,

- The User **clicks** the right button of mouse
- Train **departs** from Mysore to Bangalore.

Events include error conditions as well as normal occurrences.

For example,

- **motor jammed,**
- **transaction aborted etc..**

Types of Events

- There are **three types** of events, They are

  1. **Signal event**
  2. **Change event**
  3. **Time event**

1. Signal Event:

- A signal is an **explicit one-way transmission of information** from one object to another.
- The **signal will not expect a reply to confirm receipt**;
- A signal is a **message;** "A signal event is an occurrence of a signal in time".

Events – Signal Event...

- A signal event is the event of **sending or receiving a signal**
- Every signal event is a unique occurrence, but we can group them into signal classes. We use the **stereotype of signal in guillemets (<< signal>>)**

For example: Consider an Airline Reservation System in which there are **two signal classes**

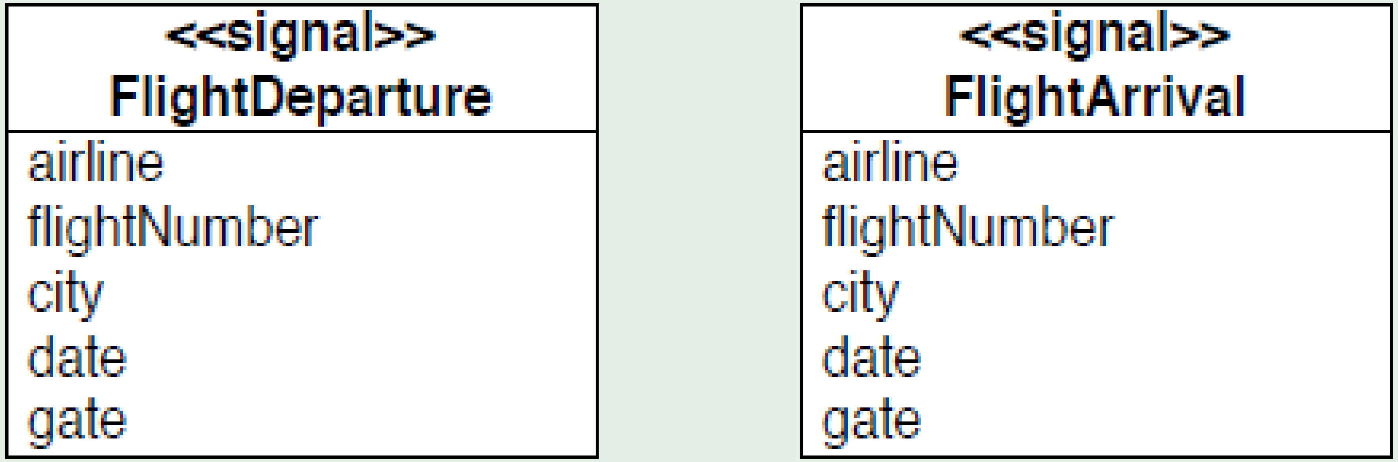
1. FlightDeparture: **airline, flightNumber, city, date, gate**
2. FlightArrival: **airline, flightNumber, city, date, gate**

Events – Signal Event…

Example

FlightDeparture: airline, flightNumber, city, date, gate

FlightArrival: airline, flightNumber, city, date, gate



2. Change Event:

- A change event is an event that is caused by the **satisfaction of a Boolean expression.**
- The intent of a change event is that the **condition is continuously tested**, and that whenever the expression **changes from false to true**, the **event happens**.

Events – Change Event…

- UML notation for a change event is the keyword "**when**" followed by a **parenthesized boolean expression**

Example

Sample change events

- when(roomTemperature > heatingSetPoint)
- when(roomTemperature < heatingSetPoint)
- when(batteryPower < lowerLimit)
- when(tirePressure < minimumPressure)

3. Time Event:

- A time event is an event caused by the occurrence of an **absolute time** or the **elapse of a time interval**

UML notation:

- For an **absolute time**, the keyword "**when**" followed by parenthesized expression involving time.
- For a **time interval**, it is the keyword "**after**" followed by a parenthesized expression that evaluates to a time duration

3. Time Event:

Example

Time event examples

- when(date = "23:58:14 January 1, 2000")
- after(1.04 seconds)

State

- A state is an abstraction of the values and links of an object.

- The UML notation for a state is a **rounded box** containing an (optional) state name.

State…

- **States** generally refer to a **verb** with a suffix "**ing**" (e.g., boiling, waiting, dialling etc.) or duration of some condition (eg. Powered, BelowFreezing),
- Whereas **events** generally refer to an **action** (e.g., StoveTurnedOn).
- Note that events may be generated by state changes (e.g., change events).

State…

Example (Some Sample States)

Example states:

- Powered
- Boiling
- InFlight
- OnApproach
- Boarding

Event Vs State

- Events represent **points in `time‘;**
- states represent **interval of time**, ie., the abstract state of an object (or collection of objects).
- Consider the switch object as an example:

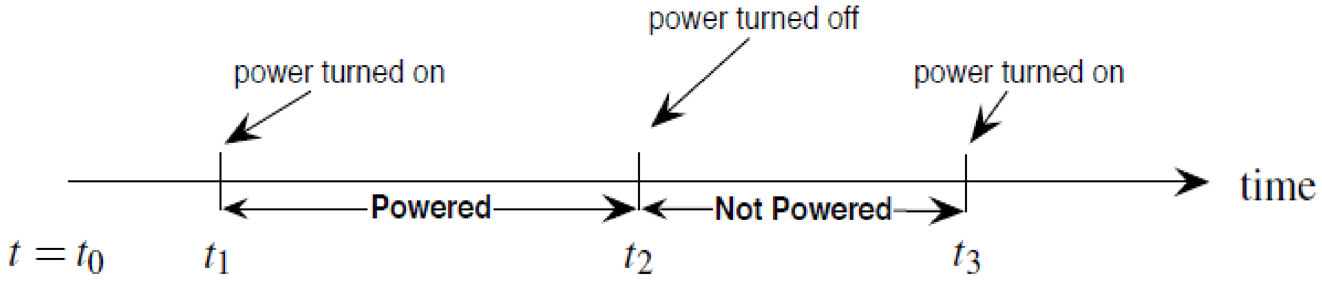


Figure: Power up, power down

Event Vs State…

- The objects in a class have finite number of possible states. Each object can only be in **one state at time.**
- Object pass through one or more states during the life time.
- A state specifies the response of an object to input events

Event Vs State…

- An events are ignored in a state, except those for which behavior is explicitly prescribed.
- Eg: if a digit is dialed in state Dial tone, the phone line drops the Dial tone & enters state **dialing**

THANK YOU

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